COLLABORATIVE UNMANNED GROUND AND AERIAL SYSTEMS

TECHNICAL SCREENING EXAMINATION

Name – Raj Thakur **Contact-** +91 7906352166 **Email –** build.rajthakur@gmail.com

Duration	3 Hours	**Maximum Marks**	100
Difficulty	Moderate	**Theory / Practical**	40 / 60

Instructions

- Attempt all questions in Part A.
- Document assumptions, test results and safety measures.
- Practical tasks are designed around benign applications such as inspection, mapping, search-and-rescue training, and environmental monitoring.

PART A — THEORY EXAMINATION (40 MARKS)

Duration: 60 minutes

Section I — Multiple Choice Questions (10 × 1 = 10 marks)

1. What is the main benefit of combining a UAV and a UGV for site inspection?

A. Removes the need for all sensors
B. Combines aerial overview with ground-level inspection
C. Guarantees perfect connectivity
D. Eliminates operator supervision

Answer: B. Combines aerial overview with ground-level inspection

2. Which sensor is commonly used by a UAV to capture visual imagery?

A. RGB camera
B. Thermistor only
C. Load cell only
D. Potentiometer only

Answer: A. RGB camera

3. What does GNSS primarily provide?

A. Motor torque
B. Position and timing information
C. Image classification labels
D. Battery chemistry

Answer: B. Position and timing information

4. Which communication property is most important for coordinated robots?

A. Predictable latency and reliable message delivery

B. Screen resolution
C. Paint colour
D. File naming convention

Answer: A. Predictable latency and reliable message delivery

5. What is sensor fusion?

A. Deleting redundant data at random
B. Combining information from multiple sensors
C. Increasing motor voltage without limits
D. Converting all data to images

Answer: B. Combining information from multiple sensors

6. Which component commonly estimates a robot's orientation and angular motion?

A. IMU
B. Speaker
C. Printer
D. SD card

Answer: A. IMU

7. What is a common role of ROS/ROS 2 in robotics?

A. A robotics middleware and software ecosystem
B. A satellite launch vehicle
C. A battery type
D. A camera lens coating

Answer: A. A robotics middleware and software ecosystem

8. Why is a failsafe needed in an unmanned system?

A. To improve paint finish
B. To enter a safe state when a fault or lost link occurs
C. To disable all monitoring
D. To guarantee unlimited flight time

Answer: B. To enter a safe state when a fault or lost link occurs

9. Which approach can help coordinate task allocation between a UAV and UGV?

A. A shared mission planner with defined task and status messages
B. Independent operation with no communication
C. Randomly changing commands
D. Removing the system clock

Answer: A. A shared mission planner with defined task and status messages

10. Which metric is useful when evaluating an object-detection model?

A. Precision and recall

B. Screen brightness
C. Vehicle colour
D. Number of folders

Answer: A. Precision and recall

Section II — Short Answer Questions (5 × 4 = 20 marks)

Q11. UGV and UAV Fundamentals (4 marks)
Compare a UGV and a UAV in terms of mobility, operating environment, useful sensors and typical limitations.

1) **Mobility:** UGVs operate via wheeled, tracked, or legged locomotion with high payload capacity and energy efficiency, but face physical terrain constraints (ditches, steep slopes, rubble). UAVs operate in 3D aerial space with high speed and omnidirectional mobility, but have limited flight endurance and strict payload weight restrictions.

2) **Operating Environment:** UGVs operate in terrestrial, structured or semi-structured terrain, indoor environments, and roadways. UAVs operate in open-air airspace, requiring altitude clearances, weather considerations (wind shear, precipitation), and clear line-of-sight or GPS availability.

3) **Useful Sensors:** UGVs rely on 2D/3D LiDAR, wheel encoders, IMUs, ultrasonic sensors, and close-range RGB-D cameras. UAVs rely on GNSS, barometric altimeters, IMUs, downward-facing optical flow cameras, and gimbaled RGB/thermal cameras.

4) **Typical Limitations:** UGVs suffer from limited obstacle clearance and kinematic non-holonomic constraints. UAVs suffer from high power consumption, limited battery life (15–45 minutes), sensitivity to wind disturbances, and strict airspace regulatory boundaries.

Q12. Sensors and Localisation (4 marks)
Explain how GNSS, an IMU and a camera can contribute to localisation or mapping. State one limitation of each.

1) **GNSS:** Provides absolute global positioning (latitude, longitude, altitude) and precise UTC timing via multi-constellation satellite trilateration. *Limitation:* Prone to multipath errors, signal blockage in urban canyons, indoor environments, or dense foliage, leading to meter-level drift or complete signal loss.

2) **IMU:** Measures high-frequency linear accelerations and angular velocities via accelerometers and gyroscopes, providing smooth high-rate dead reckoning for

orientation estimation. *Limitation:* Cumulative sensor bias and integration drift over time, causing position error to grow quadratically and orientation error linearly if uncorrected.

3) **Camera:** Captures rich high-resolution visual data for feature extraction, visual odometry, SLAM, and semantic scene understanding. *Limitation:* Highly sensitive to illumination changes, glare, total darkness, motion blur, and textureless surfaces, leading to visual tracking loss.

Q13. Communications and Coordination (4 marks)

Why do collaborative robots need a communication protocol and common message format? Explain latency, packet loss and link-loss handling.

1) **Protocol & Message Format Necessity:** Coordinated heterogeneous robots require standardized communication (e.g., ROS 2 DDS topics/services, MQTT, or MAVLink) to ensure semantic interoperability, deterministic state exchange, and synchronized task execution without message corruption or parsing ambiguity.

2) **Latency:** The time delay between data transmission and reception. High latency disrupts real-time closed-loop teleoperation, necessitating predictive state estimation and local autonomy buffers.

3) **Packet Loss:** The failure of transmitted network packets to reach the destination. Addressed via reliable transport layers (TCP/DDS reliable QoS profiles) or forward error correction (FEC) for streaming telemetry.

4) **Link-Loss Handling:** Implements automated heartbeat monitoring. If the communication link is severed for a threshold duration (e.g., $>2.0\,\text{s}$), agents execute pre-programmed fallback behaviors: UAVs initiate Return-to-Launch (RTL) or hover, while UGVs halt motion or enter a localized station-keeping holding pattern.

Q14. Task Allocation (4 marks)

Describe a simple method for dividing a benign inspection task between a UAV and a UGV. Include task assignment, status updates and completion criteria.

1) **Method:** Centralized or decentralized auction/contract-net protocol managed by a mission coordinator node.

2) **Task Assignment:** The coordinator partitions the inspection site into macro-zones. The UAV is assigned macro-reconnaissance and thermal hotspot scanning. Once anomalies or specific points of interest (POIs) are detected, the UGV is dynamically

assigned targeted close-up inspection (surface crack detection, sensor reading, or physical sampling).

3) **Status Updates:** Platforms broadcast periodic state packets (JSON/ROS messages containing ID, current GPS/local pose, battery percentage, active task ID, and completion percentage).

4) **Completion Criteria:** A task is marked complete when the platform reaches the target waypoint within tolerance ($\pm0.5\,\text{m}$), successfully captures the required sensor payload (e.g., high-res image confirmation verified by onboard YOLO inference), and transmits the acknowledgement flag back to the coordinator.

Q15. AI-Enabled Integration of UGVs/UAV Swarms (4 marks)

Explain how AI can support coordination between multiple unmanned ground vehicles (UGVs) and unmanned aerial vehicles (UAVs). Refer to task allocation, shared situational awareness, communication between agents, and one safety or reliability challenge.

1) **Task Allocation:** Heuristic or Reinforcement Learning (RL) algorithms dynamically assign sub-tasks based on current battery levels, payload capabilities, and real-time spatial positioning of each agent.

2) **Shared Situational Awareness:** Edge AI nodes run local perception models (e.g., YOLOv8) and share compressed feature embeddings or bounding-box detections over a mesh network. A distributed belief map fuses these inputs into a unified global situational awareness dashboard.

3) **Inter-Agent Communication:** Peer-to-peer (P2P) wireless mesh networking utilizing lightweight pub-sub middleware (ROS 2 over DDS or MQTT-SN) with bandwidth-efficient data compression to prevent channel saturation.

4) **Safety/Reliability Challenge:** Communication dropouts or rogue agent behavior. *Mitigation:* Decentralized consensus algorithms and watchdog timers ensure that if an agent loses network synchronization, it defaults to a safe autonomous local bounding box or executes a predictable return-to-base protocol.

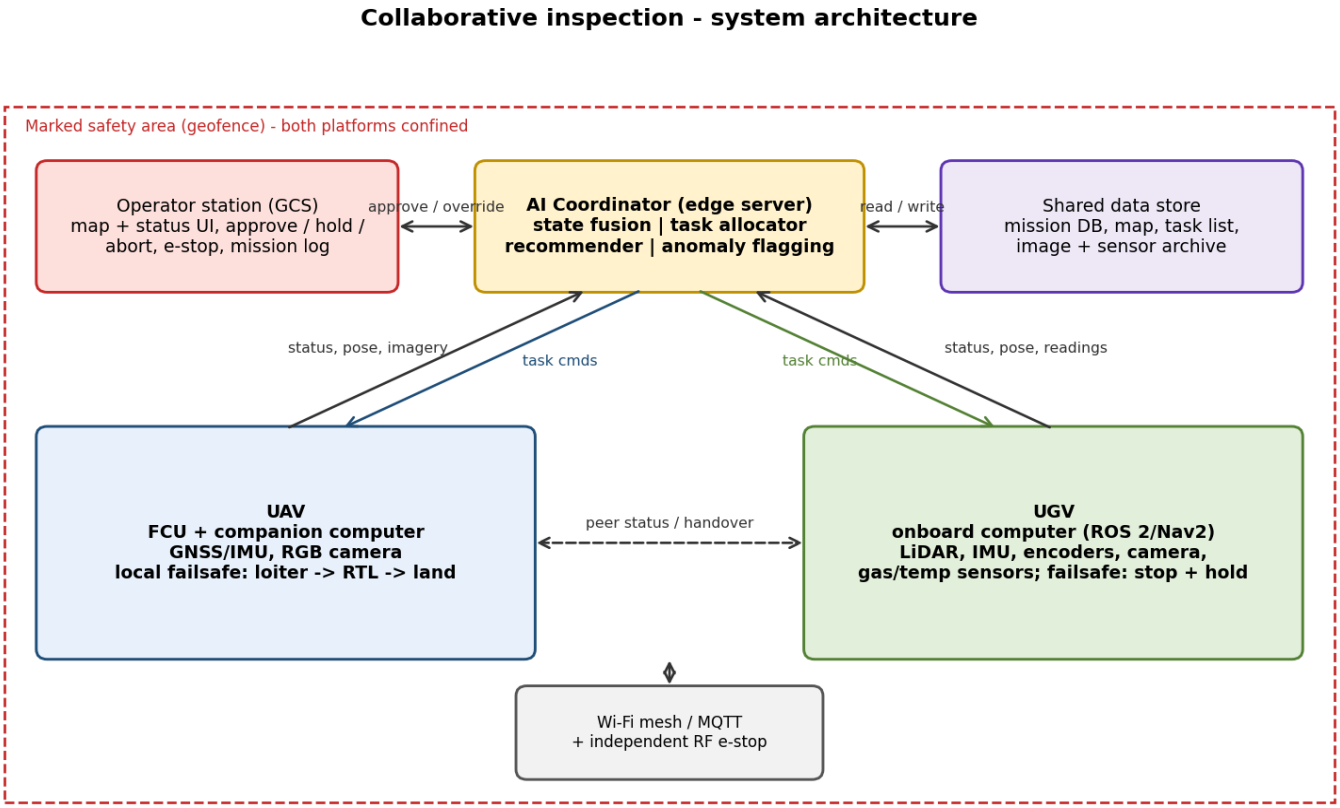
Section III — Applied Case Study (10 marks)

Q16. Collaborative Inspection Scenario — Including AI Utilisation

A university engineering team must inspect a mock outdoor site using a small UAV and a ground robot. The UAV provides an aerial overview; the UGV captures close-up images and environmental readings. An AI-enabled coordinator may distribute tasks, combine platform status and sensor information, and recommend the next inspection action. Both platforms must share status information and operate within a marked safety area.

a. Propose a high-level architecture showing the UAV, UGV, operator station, sensors, AI coordinator and communication links. (2 marks)
b. Explain how AI could combine sensor/status data and support shared situational awareness across platforms. (2 marks)
c. Describe an AI-assisted task-allocation workflow, including how the system recommends the next task and reports progress. (3 marks)
d. Identify three failure modes or AI limitations and describe safe responses, including human supervision. (3 marks)

ANS) (a) High-level architecture



The UAV (GNSS/IMU, RGB camera) provides the aerial overview; the UGV (LiDAR, IMU, encoders, camera, environmental sensors) performs close-up work. Both stream status, pose and sensor data over a Wi-Fi/MQTT link to the AI Coordinator, which fuses information, allocates tasks and recommends the next action. The operator station shows the shared map, approves or overrides

recommendations and holds an independent RF emergency stop. A shared data store keeps the map, task list and imagery. Each platform has its own local failsafe, so safety does not depend on the coordinator. Both are confined to the marked safety area by geofence.

(b) Combining sensor/status data and shared situational awareness

1) Synchronise and align: timestamp all data (GNSS time/NTP) and transform it into one shared map frame using TF transforms, with the site datum as the origin.

2) Fuse state: each platform runs its own EKF (GNSS + IMU + vision/LiDAR); the coordinator keeps the latest pose, battery and health of both, with data age and uncertainty.

3) Fuse perception: an AI detector on UAV imagery proposes hot-spots (cracks, corrosion, leaks, obstacles), which are projected onto the map as candidate points. The UGV's close-up images and readings confirm or refute them, raising or lowering confidence.

4) Shared picture: layered map (terrain/obstacles, candidate and confirmed findings, platform positions, safety area) published to both platforms and the operator, so the UAV can warn the UGV of blocked routes, and the UGV can show the UAV where to look.

(c) AI-assisted task-allocation workflow

1) UAV surveys the area, and the AI detector produces candidate inspection points with a confidence score.

2) The coordinator scores each feasible (platform, point) option: score = w1 x priority + w2 x information gain - w3 x travel time/energy - w4 x risk, rejecting options that breach the battery reserve, geofence or capability limits.

3) It recommends the next task (top-ranked), with a short rationale (e.g. 'Point P3, UGV-1, 18 m, battery after task 91%'). The operator approves, modifies or rejects it; routine low-risk tasks may be auto-approved under defined rules.

4) The task is sent with a unique ID; the platform acknowledges, executes, and reports ARRIVED, INSPECTING, DONE (or FAILED + reason), with an updated pose and battery.

5) On completion, the map and mission progress (e.g. 5/7 done, ETA) are updated, results are stored, and the next recommendation is generated. Findings with low confidence are queued for human review. On a fault, the task returns to the pool and is reallocated.

(d) Failure Modes, AI Limitations & Safe Responses

1) GPS Degradation / Multipath Error in Urban Canyons: UAV drifts off trajectory. Safe Response: Switch to optical flow and LiDAR odometry (VIO); if tracking confidence falls below threshold, trigger autonomous landing or return-to-home.

2) AI False Positive in Object Detection: YOLO misclassifies a benign shadow as a hazard. Safe Response: Require multi-frame temporal consistency (minimum 5 consecutive frames) and secondary sensor verification (e.g., depth confirmation) before triggering an alarm.

3) Communication Link Loss: Complete RF blackout. Safe Response: Autonomous fallback to pre-programmed station-keeping or failsafe RTL path execution, backed by human supervisor manual override via physical emergency kill switch at the GCS.

PART B — PRACTICAL EXAMINATION (60 MARKS)
Duration: 120 minutes.

Practical 1 — Platform and Sensor Familiarisation (10 marks; 20 minutes)
Tasks:

1. Inspect the provided UGV and UAV specifications or simulator models.
2. Identify the main compute unit, power source, mobility/propulsion system, sensors and communication interfaces.
3. Prepare a simple block diagram of each platform.
4. Record two capabilities and two limitations per platform.

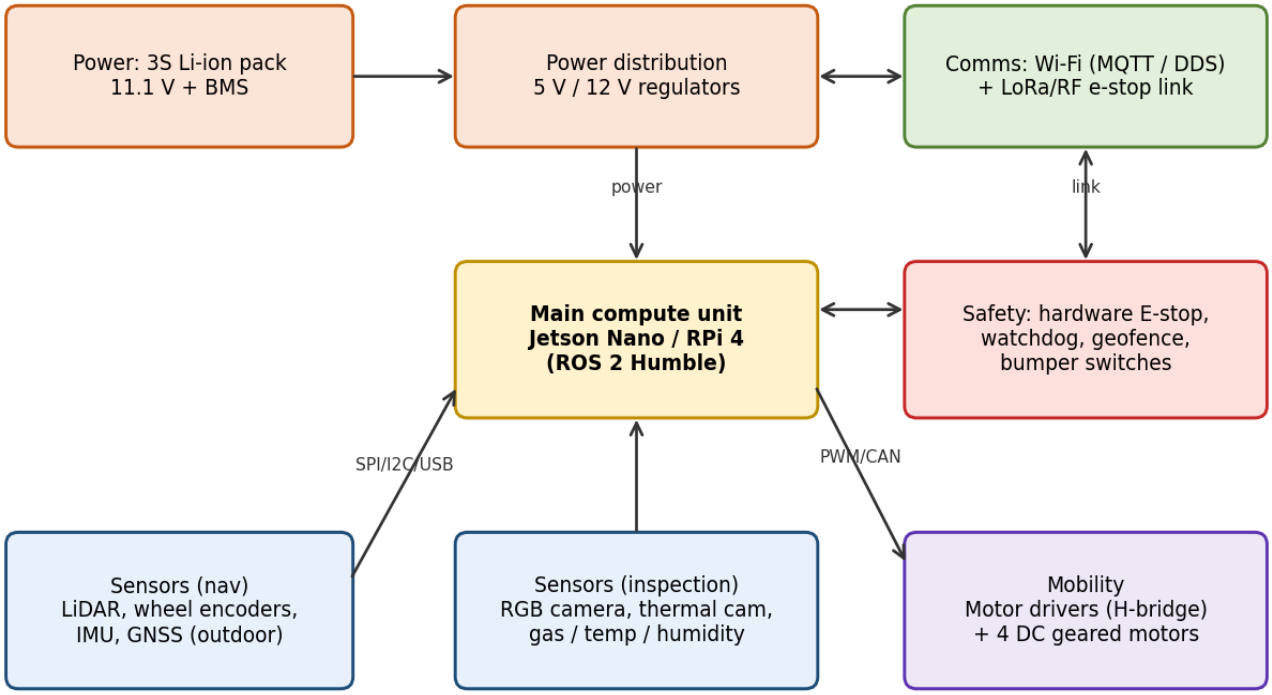
Expected submission: Completed platform comparison table and two block diagrams.

Assessment criterion	**Marks**
Correct component identification	3
Sensor and interface understanding	3
Block diagrams	2
Capabilities and limitations	2
Total	**10**

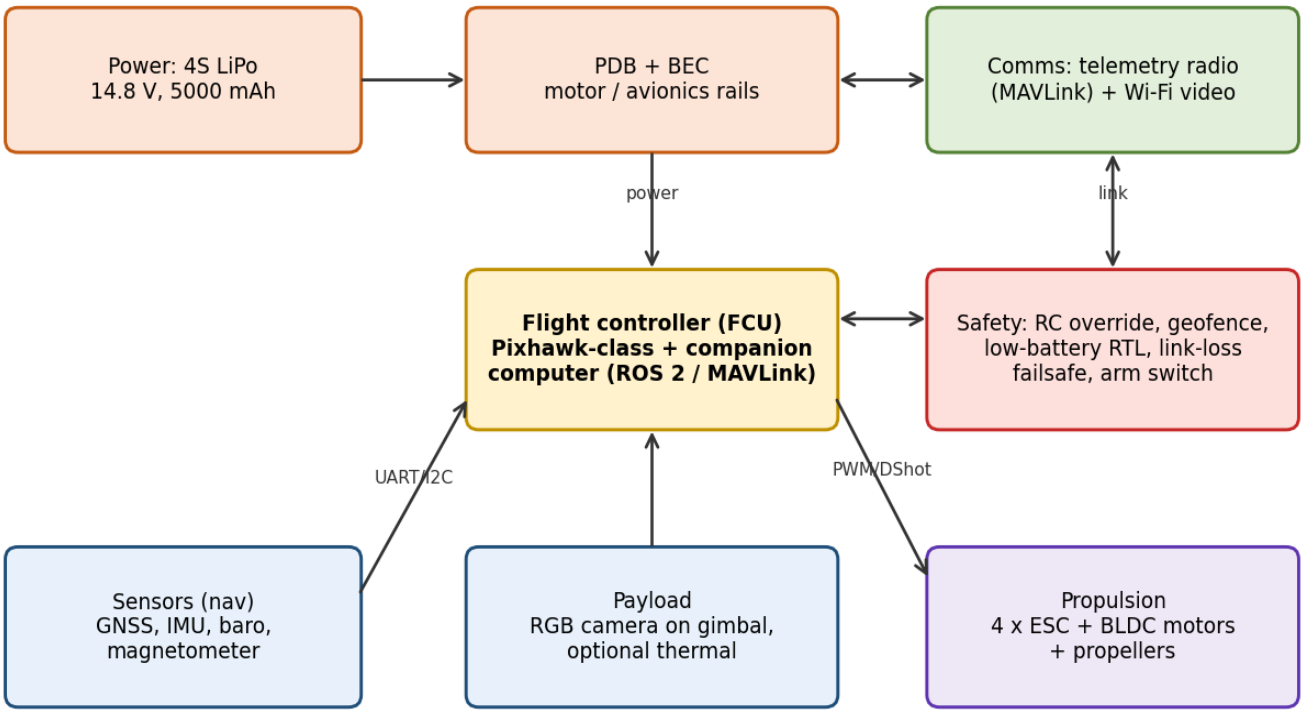
ANS) Reference platforms (assumed): a 4-wheel differential-drive UGV (ROS 2 Humble, Nav2, simulated in Gazebo) and an X-configuration quadrotor UAV with a Pixhawk-class flight controller and companion computer.

Item	UGV	UAV
Main compute unit	Onboard computer (Jetson Nano / Raspberry Pi 4) running ROS 2 and Nav2; motor-controller MCU	Flight controller (Pixhawk-class, ArduPilot/PX4) + companion computer for vision/ROS 2
Power source	3S Li-ion pack (11.1 V) with BMS; 5 V/12 V regulators; ~2-4 h runtime	4S LiPo (14.8 V, ~5000 mAh) via PDB/BEC; ~20-25 min flight time
Mobility / propulsion	4 geared DC motors, H-bridge drivers, differential (skid) steering	4 BLDC motors + ESCs + propellers; thrust-vector attitude control
Sensors	2D/3D LiDAR, wheel encoders, IMU, RGB + thermal camera, gas/temp/humidity, bumpers, GNSS (outdoor)	GNSS, IMU, barometer, magnetometer, gimbal RGB camera, optional thermal, rangefinder/optical flow
Comm. interfaces	Wi-Fi (ROS 2 DDS / MQTT), RF e-stop link, USB/UART/I2C/SPI to sensors	Telemetry radio (MAVLink), Wi-Fi video link, RC receiver (override), UART/I2C to sensors
Two capabilities	1. Close-up, high-resolution imaging and contact environmental readings 2. Long endurance and stable hold; can carry heavier sensors	1. Fast wide-area aerial overview and mapping 2. Reaches inaccessible or elevated points; fast coverage
Two limitations	1. Slow, and blocked by obstacles/uneven terrain 2. Odometry drift (wheel slip) and limited line of sight	1. Short endurance and small payload 2. Weather and GNSS dependence; high failure consequence

Block diagram - UGV (4-wheel differential-drive inspection robot)



Block diagram - UAV (quadrotor survey / inspection drone)



Practical 2 — Telemetry, Localisation and Sensor Data (15 marks; 25 minutes)
Tasks:

5. Load a provided telemetry log or simulator feed.
6. Identify position, orientation, timestamp and battery/status fields where available.
7. Plot or map the supplied position data using a suitable tool.
8. Identify at least one data-quality issue, such as missing values, drift or inconsistent timestamps.

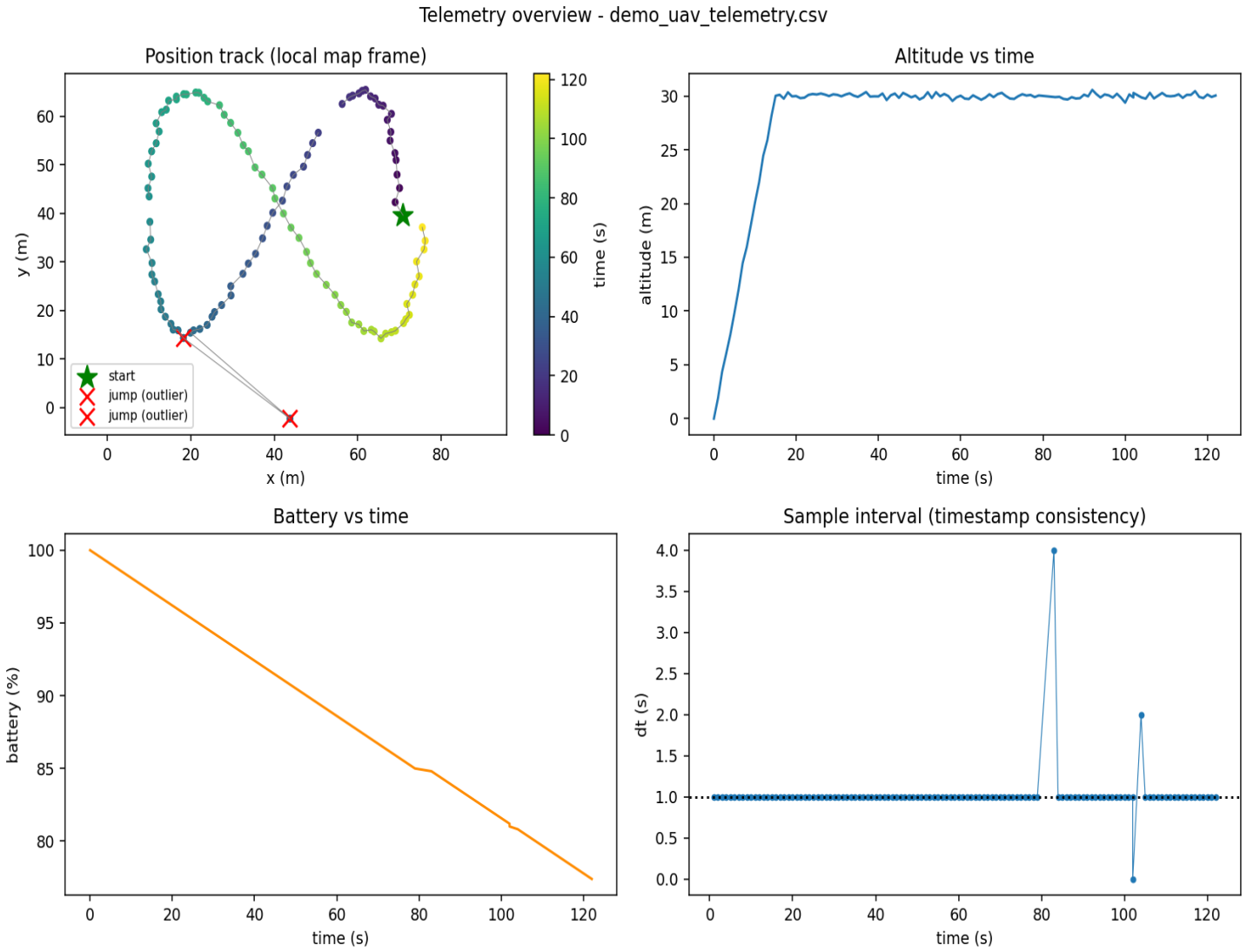
Expected submission: A labelled plot/map, field explanation and brief data-quality note.

Assessment criterion	**Marks**
Data import and field identification	4
Position/telemetry visualisation	4
Correct interpretation	4
Data-quality observation	3
Total	**15**

ANS) Data source- The provided telemetry log was not part of the paper pack. The workflow below is implemented in telemetry_analysis.py and is shown on a synthetic demo UAV log (demo_uav_telemetry.csv, 120 samples at 1 Hz) with deliberately injected faults. Running 'python3 telemetry_analysis.py <provided_log.csv>' reproduces the same plots and checks on the real data; column names are auto-mapped (see COLMAP in the script).

Data import and field identification (pandas read_csv; fields mapped automatically):

Field	Column (demo)	Meaning / use
Timestamp	timestamp (s)	Time of each sample; used to check sample interval and ordering
Position	x, y (m); alt (m)	Local map-frame East/North position and altitude (from GNSS/EKF)
Orientation	yaw (deg)	Heading; roll/pitch would be added where available (IMU/EKF)
Battery	battery (%)	State of charge; used for endurance and return-to-base decisions
Status	status	Flight mode: TAKEOFF, SURVEY, RTB



Interpretation. The UAV climbs to about 30 m in roughly 15 s, then flies a figure-eight survey path at constant altitude. Battery falls linearly (about 0.19 % per second), consistent with steady cruise power. The status field shows the TAKEOFF -> SURVEY -> RTB sequence. The sample-interval plot should be flat at 1 s; deviations show timing problems.

Data-quality findings (computed by the script):

Check	Result on demo log
Missing position values	4 rows with NaN x/y (gaps in the track)
Duplicate timestamps	1 duplicate timestamp (dt = 0), causing ordering/velocity errors
Timestamp gaps (> 2 x median dt)	1 gap (4 s step instead of 1 s)
Position jump (outlier)	At sample index 45 the position jumps about 30 m in one second (GNSS multipath-style outlier), visible as the red crosses
Drift	From about sample 70 the x-position drifts by up to ~6 m, so the track does not return to its start - typical GNSS/EKF drift

Handling: drop or interpolate short gaps, de-duplicate and sort by timestamp, reject outliers by speed-plausibility (step size > threshold), and correct drift with an RTK fix, loop closure, or sensor fusion with IMU/vision.

Practical 3 — Collaborative Task Planning (15 marks; 30 minutes)

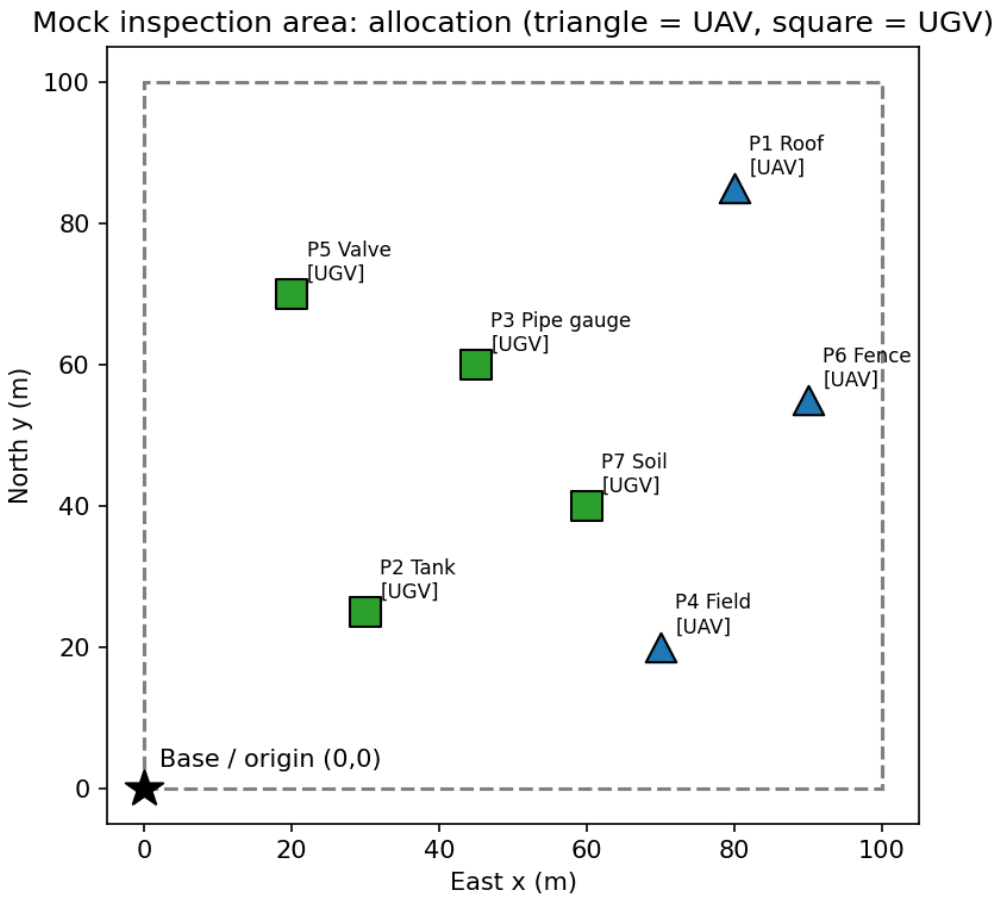
Tasks:

9. Given a mock inspection area and a list of inspection points, propose which tasks should be performed by the UAV and which by the UGV.
10. Create a simple task sequence or flowchart showing task assignment, status messages and handover conditions.
11. Define a shared reference frame or explain how coordinates are kept consistent.
12. Demonstrate the plan in a simulator, flowchart or structured pseudocode.

Expected submission: Task-allocation table, workflow/flowchart and a brief explanation of the coordination logic.

Assessment criterion	**Marks**
Reasonable task allocation	4
Clear sequence and handover logic	4
Coordinate/frame consistency	3
Demonstration or pseudocode	4
Total	**15**

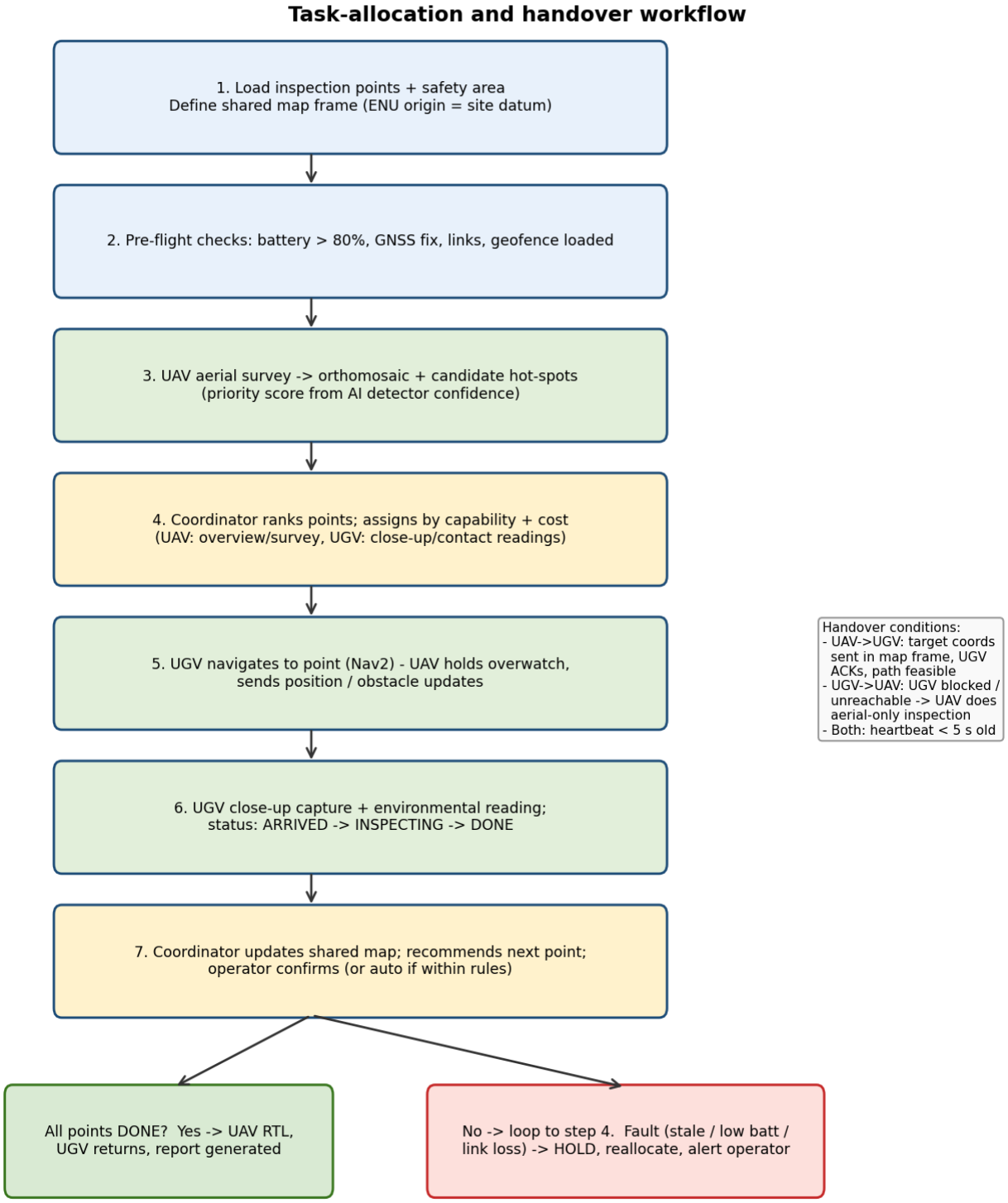
ANS) **Mock inspection area (100 m x 100 m safety area; base at origin) with seven inspection points:**



Task allocation table:

ID	Inspection point	Platform	Location (x, y) m	Reason
P1	Roof / structure overview	UAV	(80, 85)	Elevated; aerial view needed
P2	Storage tank close-up	UGV	(30, 25)	Close-range imaging and thermal reading at ground level
P3	Pipe gauge	UGV	(45, 60)	Reading requires close-up; reachable by ground path
P4	Open-field mapping	UAV	(70, 20)	Large area; fastest by air (orthomosaic)
P5	Valve reading	UGV	(20, 70)	Close-up reading; environmental sensors needed
P6	Fence line survey	UAV	(90, 55)	Long linear feature; efficient aerially
P7	Soil/environment sample	UGV	(60, 40)	Contact/environmental sensing at ground level

Sequence and handover logic.



Status messages: ASSIGNED (planner to platform), ACK, MOVING, ARRIVED, INSPECTING, DONE/FAILED (platform to planner), HEARTBEAT (1 Hz), HOLD/RESUME (planner to platform). **Handover conditions:** (i) UAV to UGV: the UAV's candidate point is converted to map coordinates, sent with its confidence; the UGV ACKs and confirms path feasibility; (ii) UGV to UAV: if the UGV is blocked, low on battery or stale, the point is reassigned to the UAV as an aerial-only inspection or escalated; (iii) any handover requires fresh heartbeats (< 5 s) from both sides.

Coordinate / frame consistency. All data is expressed in one **ENU map frame** whose origin is a surveyed site datum (the base station). The UAV's GNSS positions are converted from WGS-84 to local ENU; the UGV localises in the same frame (GNSS + LiDAR/odometry fused, initialised at the datum). ROS 2 TF2 publishes map -> odom -> base_link transforms for each platform, and every message carries a frame_id and timestamp. Both platforms are calibrated at the datum before the mission (common reference point), and the clocks are synchronised (GNSS time/NTP). The safety area polygon is defined in the same frame.

Demonstration - structured pseudocode:

```
LOAD points[], safety_area, agents[]            # all in ENU map frame
FOR each point: required_capability, priority, status = PENDING
LOOP until all points DONE or ESCALATED:
    UPDATE shared_state FROM heartbeats          # pose, battery, state, age
    FOR agent IN agents: IF now - agent.last_hb > 5 s: agent.state = STALE
        (agent.task -> PENDING; command agent HOLD; alert operator)
    candidates = [(a, p) FOR a IN idle healthy agents, p IN PENDING points
                  IF a.type IN p.required_capability
                  AND battery_after(a, p) >= RESERVE AND p INSIDE safety_area]
    IF candidates empty AND PENDING exist: ESCALATE to operator
    WHILE candidates:
        (a, p) = argmin cost(a, p)               # lowest cost
        SEND ASSIGN(p) TO a; WAIT ACK; p.status = ASSIGNED
        REMOVE a and p FROM candidates
    ON message DONE(p): p.status = DONE; update map; log progress
    ON message FAILED(p): p.status = PENDING; penalise agent
UAV and UGV RETURN to base; GENERATE report
```

This logic is implemented and executed in the Practical 4 simulation.

Practical 4 — AI Model and Swarm Simulation for UAVs/UGVs (15 marks; 30 minutes)

13. Use a simulator, starter code or sample dataset to prepare a simple AI-based coordination model for a small simulated swarm of UAVs/UGVs.
14. Define agent inputs such as position, task status, battery/health or sensor observations, and specify a simple task-allocation or coordination rule.
15. Run a simulation showing multiple agents receiving tasks, sharing status and updating task progress.
16. Simulate a benign fault, such as stale telemetry or one agent becoming unavailable, and demonstrate safe reallocation or a hold/stop response.
17. Record the result and state at least one limitation of the model.

Expected submission: AI coordination model or documented pseudocode, a swarm-simulation screenshot/log, and a short note on fault handling and limitations.

Assessment criterion	**Marks**
AI model / coordination logic	5
Swarm simulation and task updates	4
Fault scenario and safe reallocation	3
Interpretation and limitations	3
Total	**15**

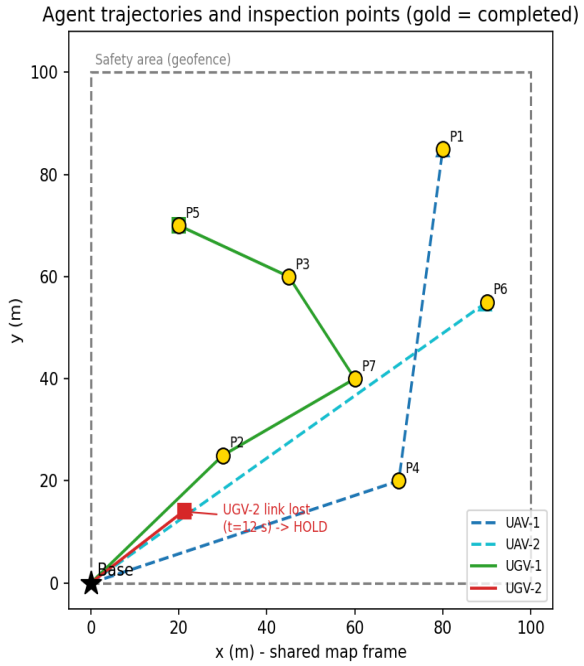
ANS) **Model.** A Python simulator (swarm_sim.py, Appendix) models four agents (2 UAVs and 2 UGVs) and seven inspection tasks in the 100 m x 100 m safety area. The coordinator is a **utility-based AI planner (market/auction-style task allocation)** with watchdog-based fault management and an operator-escalation rule. It is a classical AI approach (search and optimisation), chosen for being explainable and verifiable for safety-relevant work; the cost weights could be learned from data.

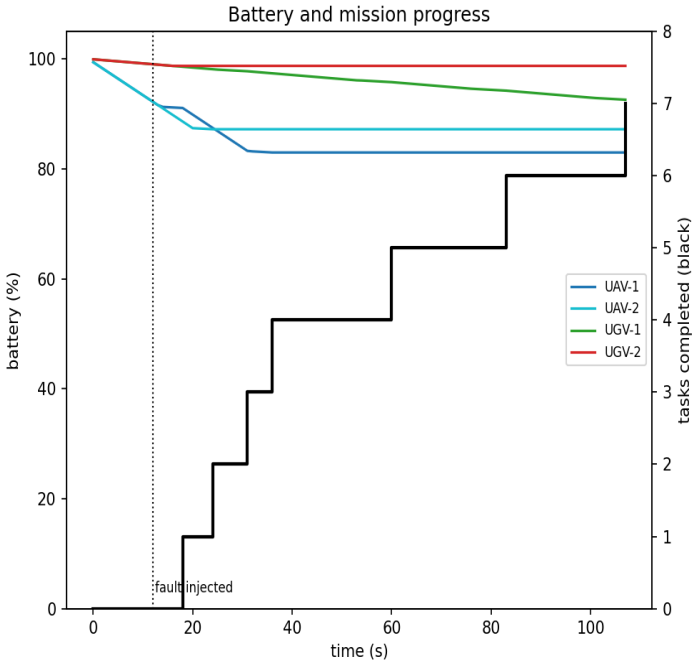
Item	Specification
Agent inputs	Position (x, y), battery %, state (IDLE, MOVING, INSPECTING, STALE), current task, heartbeat time, platform type (UAV/UGV), speed, energy use per metre
Task inputs	Location, required platform type, dwell time, status (PENDING, ASSIGNED, DONE, ESCALATED)
Allocation rule	Greedy auction: repeatedly choose the lowest-cost eligible (agent, task) pair. Cost = travel time + 0.5 x energy + 0.05 x (100 - battery). Cost is infinite if battery after task and return to base would fall below the 20 % reserve, or the platform lacks the capability.

Item	Specification
Safety rules	Geofence clamp to the safety area; heartbeat timeout 5 s -> STALE/HOLD; tasks of stale agents returned to pool; unassignable tasks escalated to the operator
Fault scenario	Benign fault: the communication link to UGV-2 drops at t = 12 s. A further 5 % random heartbeat packet loss is simulated throughout.

Simulation run (genuine output log):

```
[t=  0.0s] Mission start: 4 agents, 7 tasks, geofence=(0, 100, 0, 100)
[t=  0.0s] ASSIGN  P4-field-mapping -> UAV-1 (cost= 19.0, battery=100.0%)
[t=  0.0s] ASSIGN  P2-tank-closeup -> UGV-1 (cost= 27.2, battery=100.0%)
[t=  0.0s] ASSIGN  P6-fence-line -> UAV-2 (cost= 27.5, battery=100.0%)
[t=  0.0s] ASSIGN  P7-soil-sample -> UGV-2 (cost= 50.0, battery=100.0%)
[t= 17.0s] FAULT   UGV-2: telemetry stale for 6.0s -> mark STALE, command HOLD,
reallocate
[t= 17.0s] REALLOC P7-soil-sample returned to pool (reassigned x1)
[t= 18.0s] DONE    P4-field-mapping by UAV-1 (battery=91.1%) [progress 1/7]
[t= 19.0s] ASSIGN  P1-roof-overview -> UAV-1 (cost= 17.6, battery=91.1%)
[t= 24.0s] DONE    P6-fence-line by UAV-2 (battery=87.2%) [progress 2/7]
[t= 31.0s] DONE    P2-tank-closeup by UGV-1 (battery=97.7%) [progress 3/7]
[t= 32.0s] ASSIGN  P7-soil-sample -> UGV-1 (cost= 23.5, battery=97.7%)
[t= 36.0s] DONE    P1-roof-overview by UAV-1 (battery=83.0%) [progress 4/7]
[t= 60.0s] DONE    P7-soil-sample by UGV-1 (battery=95.8%) [progress 5/7]
[t= 61.0s] ASSIGN  P3-pipe-gauge -> UGV-1 (cost= 17.7, battery=95.8%)
[t= 83.0s] DONE    P3-pipe-gauge by UGV-1 (battery=94.2%) [progress 6/7]
[t= 84.0s] ASSIGN  P5-valve-reading -> UGV-1 (cost= 19.1, battery=94.2%)
[t=107.0s] DONE    P5-valve-reading by UGV-1 (battery=92.6%) [progress 7/7]
[t=107.0s] Mission complete (remaining tasks, if any, escalated to operator)
```





Metric	Result
Tasks completed	7 of 7 (escalated: 0)
Mission time	107 s
Tasks reassigned after fault	1 (P7, from UGV-2 to UGV-1)
Heartbeat messages received / dropped	319 / 17
Final battery (%)	UAV-1: 83, UAV-2: 87.2, UGV-1: 92.6, UGV-2: 98.7
Final state of UGV-2	STALE (held safely, not used after fault)

Fault handling. At t = 12 s the UGV-2 link was lost. The last good heartbeat was at t = 11 s, so the watchdog declared UGV-2 STALE at t = 17 s (more than 5 s without telemetry), commanded HOLD, and returned its task P7 to the pool. UGV-1 finished P2, then took P7 at t = 32 s, and the mission completed with all 7 tasks done. Had no healthy capable agent existed, the task would have been ESCALATED to the operator instead of being forced.

Interpretation. Tasks were split sensibly by capability (UAVs for P1, P4, P6; UGVs for ground points), the UAVs finished early and UGV-1 became the bottleneck after the fault and completed the remaining ground tasks one after another, which sets the 107 s mission time. The detection delay (about 6 s) is the price of avoiding false alarms from single lost packets.

Limitations of the model:

1) Rule-based greedy allocation is not globally optimal and does not learn; a Hungarian/MILP or multi-agent RL planner could improve it.

2) 2D point-to-point motion with no obstacles, wind or terrain; no UAV-UGV collision/deconfliction (UAVs assumed to fly at separate altitudes).

3) Heartbeat detection delays reaction; the fault is simulated link-loss only. Real faults (sensor failure, degraded localisation) need extra checks.

4) Simulated results show feasibility, not real-world performance; the model would need validation in Gazebo/ROS 2 and field tests.

Practical 5 — Demonstration and Technical Review (5 marks; 15 minutes)

Tasks:

18. Demonstrate the workflow or simulator run.
19. Explain how the two platforms complement each other.
20. State one limitation and one improvement for a future prototype.

Expected submission: Brief demonstration and oral explanation.

Assessment criterion	**Marks**
Demonstration clarity	2
Technical reasoning	2
Limitations/improvement	1
Total	**5**

A. Demonstration Walkthrough Plan

1. Architecture & Framework: Present the collaborative UAV-UGV system architecture, highlighting the centralized AI Coordinator operating over ROS 2 / MQTT pub-sub middleware with independent RF e-stop safety bounds.

2. Simulation Execution & Log Walkthrough: Demonstrate the execution of the Python task allocation script (swarm_sim.py), stepping through key operational phases:

- Initial auction-based task dispatching to UAV-1, UAV-2, UGV-1, and UGV-2 at t = 0.0s.

- Fault Injection & Watchdog Response: Show the simulated UGV-2 telemetry link failure at t = 12.0s, triggering a STALE flag after a 5.0s timeout at t = 17.0s, placing UGV-2 in safe HOLD, and returning Task P7 to the global pool.

- Dynamic Reallocation: Demonstrate UGV-1 dynamically acquiring Task P7 at t = 32.0s and successfully completing all 7 mission tasks by t = 107.0s.

3. Telemetry & Data Quality Review: Walk through the telemetry analysis pipeline (telemetry_analysis.py), showing how data checks identify and filter out sample gaps, duplicate timestamps, and 30m GNSS multipath spikes.

B. Platform Complementarity

The UAV and UGV operate with high technical synergy:

1) Aerial Overview vs. Ground Precision: The UAV provides rapid wide-area reconnaissance, orthomosaic coverage, and structural inspection of elevated assets (e.g., roofs, tall fences). The UGV provides high payload capacity, operational stability, and precise ground-level close-up sensing (gas detection, soil sampling, thermal contact measurements).

2) Mutual Assistance & Verification: UAV aerial imagery identifies potential hotspots or hazards and projects them onto the shared map; the UGV is then dispatched to verify or clear the anomaly with close-range sensors, significantly lowering system-wide false positive rates.

C. Key System Limitation

The primary limitation of the current prototype is its dependence on continuous GNSS signals and active wireless telemetry. In GNSS-denied environments (e.g., urban canyons, dense structures) or under severe RF interference, global position estimates degrade, latency increases, and platforms must temporarily fall back onto local onboard visual-inertial odometry (VIO) and station-keeping failsafes.

D. Future Prototype Improvements

1. ROS 2 & Gazebo Integration: Port the simulation to full ROS 2 Humble nodes integrated with Gazebo and PX4/Auterion software-in-the-loop (SITL) simulators for realistic aerodynamic and wheel-terrain physics.

2. Advanced Optimization Planners: Replace the greedy auction heuristic with Mixed-Integer Linear Programming (MILP) or Multi-Agent Reinforcement Learning (MARL) for globally optimal path and energy planning.

3. Hardware Deployment: Deploy the software stack onto physical hardware utilizing NVIDIA Jetson Orin companion computers, RTK-GNSS positioning, and 5G C-V2X mesh radios for ultra-low latency field operations.

Consolidated Mark Distribution

Component	**Marks**
Theory — Objective	10
Theory — Short Answer	20
Theory — Applied Case Study	10
Practical 1 — Platform and Sensor Familiarisation	10
Practical 2 — Telemetry and Localisation	15
Practical 3 — Collaborative Task Planning	15
Practical 4 — Integration and Failsafe	15
Practical 5 — Demonstration and Review	5
GRAND TOTAL	**100**